Project Design Phase-II Data Flow Diagram & User Stories

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Team ID	
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Section 1: Data Flow Diagrams

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Shape / Symbol	Component Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Clinician, Patient, EHR System).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., 1.0 Ingest Clinical Data, 2.0 Calculate Risk Scores, 3.0 Generate Visual Analytics).
Rectangle (solid) / Open Rectangle	**Data Store**	A place where data is held for later use (e.g., Patient Clinical Repository, Risk Metrics Datastore).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed (e.g., Raw Vitals,

Shape / Symbol	Component Name	Description
		Processed Risk Scores, Visual Queries).

Section 2: Health360 DFD Level 0 (Context Diagram) & Level 1 Architecture

Level 0 Context Diagram Specification

- **External Entities:**
  - Healthcare Clinician
  - Patient EHR Intake System
  - Public Health Administrator
- **Central Process:**
  - **0.0 Health360 Clinical Risk Analytics System**
- **Data Flows:**
  - **Inflow:** Ingestion of 5,000 Patient Records containing vital and lifestyle attributes (Blood Pressure, Fatigue, Exercise frequency, Substance habits).
  - **Outflow:** Delivery of Interactive Diagnostic Visuals, 5-Point Clinical Stories, and High-Risk Preventive Alerts to decision-makers.

Level 1 Process Breakdown

- **Process 1.0: Patient Data Ingestion & Validation**
  - Ingests raw clinical CSV/EHR data containing vital signs and lifestyle logs.
  - Validates and cleans records for missing values across demographic cohorts.
  - Stores structured records in the **Patient Clinical Repository**.
- **Process 2.0: Multi-Factor Clinical Risk Scoring & Correlation Engine**
  - Reads validated data from the **Patient Clinical Repository**.
  - Computes blood pressure cohorts, fatigue incidence rates, and combined multi-variable lifestyle risk scores.
  - Stores normalized calculated metrics into the **Clinical Metrics Store**.
- **Process 3.0: Visual Analytics & Dashboard Rendering**
  - Queries calculated metrics from the **Clinical Metrics Store**.
  - Renders 6 distinct interactive visual widgets: BP Bar Chart, Age Distribution Pie Chart, Fatigue Stacked Bar Chart, Exercise Donut Chart, Substance Frequency Bar Chart, and Exercise vs. Age Risk Heatmap.
- **Process 4.0: Risk Alert & Story Reporting**
  - Synthesizes metrics into structured multi-tab visual presentations.
  - Emits automated preventive risk alerts and actionable 5-point clinical data stories targeted at Healthcare Clinicians and Administrators.

Section 3: User Stories Table

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Clinical Risk Analyst	Data Ingestion & Profiling	USN-01	As a Clinical Risk Analyst, I want to import and validate 5,000 patient records across 6 age groups so that the dataset is clean for vital analysis.	All 5,000 records loaded with zero missing demographic or BP values.	High	Sprint-1
Clinical Risk Analyst	Hypertension Monitoring	USN-02	As a Clinical Risk Analyst, I want to visualize average systolic blood pressure across age demographics so that prehypertensive cohorts are detected early.	Clear bar chart displaying average BP values (Mean ~134.4 mmHg) across all 6 age cohorts.	High	Sprint-1
Healthcare Clinician	Symptom & Fatigue Analysis	USN-03	As a Clinician, I want to analyze fatigue condition incidence against patient age categories so that I can diagnose chronic fatigue patterns.	Stacked bar chart showing binary fatigue breakdown (50.8% overall prevalence) per demographic.	High	Sprint-1
Lifestyle Specialist	Habit & Substance Correlation	USN-04	As a Lifestyle Specialist, I want to compare alcohol and smoking consumption frequencies across age brackets so that targeted cessation programs can be recommended.	Grouped bar chart accurately displaying substance use counts across cohorts.	Medium	Sprint-2
Public Health Officer	Predictive Risk Heatmap	USN-05	As a Public Health Officer, I want to view an exercise level vs age risk score heatmap so that multi-variable lifestyle dependencies are easily evaluated.	6×3 matrix heatmap showing normalized risk scores (Baseline 100.0) across physical activity tiers.	High	Sprint-2
Healthcare Administrator	Executive Data Story	USN-06	As an Administrator, I want a 5-point guided clinical data story presentation so that executive teams can make immediate preventive healthcare policies.	Seamless multi-tab data story reviewing macro metrics, BP variance, and clinical recommendations.	Medium	Sprint-2
Web User / Clinician	Web Dashboard Access	USN-07	As a Web User, I want a responsive web portal hosted on GitHub Pages so that the clinical	Web dashboard loads seamlessly with interactive Chart.js graphs and KPI cards.	High	Sprint-1

	User Story Number	User Story / Task	Acceptance Criteria	Priority
		dashboard is accessible across mobile and desktop.		